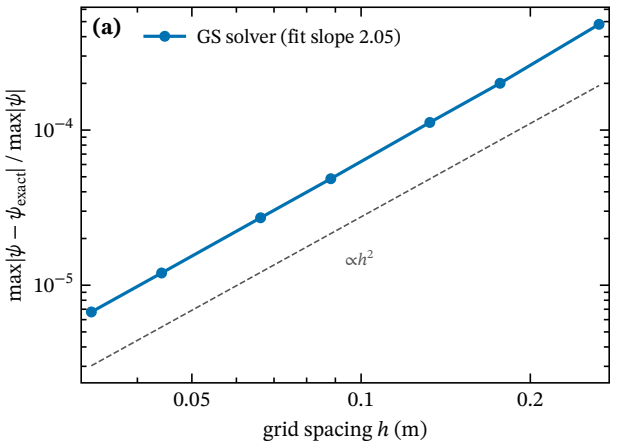
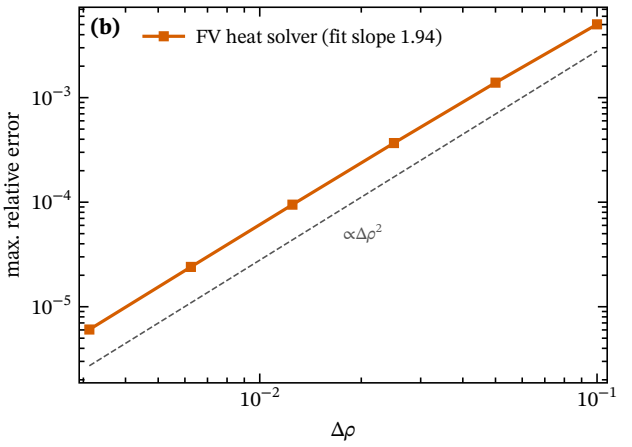
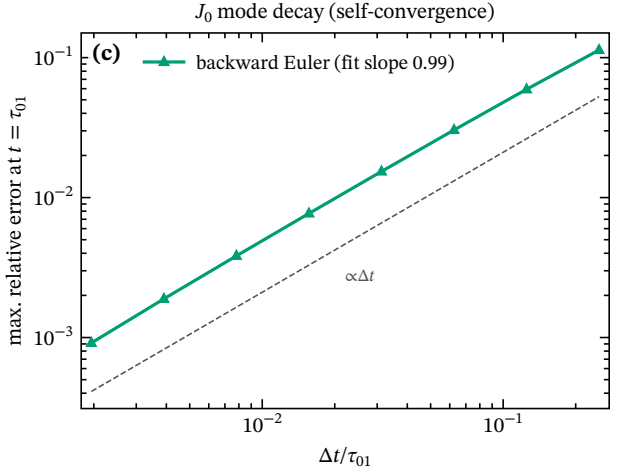
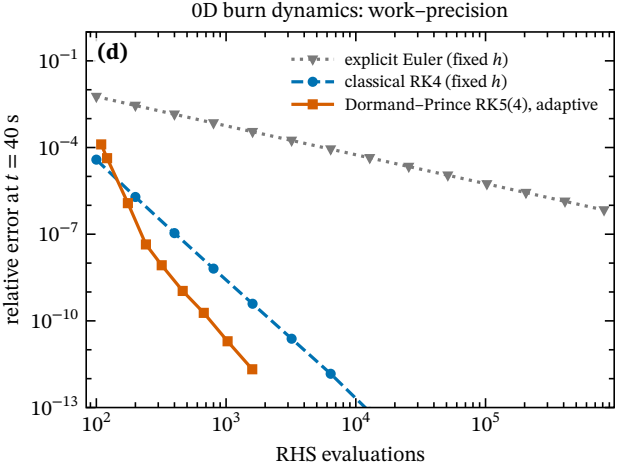

Grad–Shafranov vs Solov'ev
(a)
GS solver (fit slope 2.05)
$\max|\psi - \psi_{\mathrm{exact}}| \,/\, \max|\psi|$
10^{-4}
10^{-5}
$\propto h^2$
0.05
0.1
0.2
grid spacing h (m)


Steady diffusion, uniform source
(b)
FV heat solver (fit slope 1.94)
max. relative error
10^{-3}
10^{-4}
10^{-5}
$\propto \Delta\rho^2$
10^{-2}
10^{-1}
$\Delta\rho$


J_0 mode decay (self-convergence)
(c)
backward Euler (fit slope 0.99)
max. relative error at $t = \tau_{01}$
10^{-1}
10^{-2}
10^{-3}
$\propto \Delta t$
10^{-2}
10^{-1}
$\Delta t/\tau_{01}$


0D burn dynamics: work–precision
(d)
explicit Euler (fixed h)
classical RK4 (fixed h)
Dormand–Prince RK5(4), adaptive
relative error at $t = 40$ s
10^{-1}
10^{-4}
10^{-7}
10^{-10}
10^{-13}
10^{2}
10^{3}
10^{4}
10^{5}
RHS evaluations